

MedGraphRAG: A Safety-First Hybrid Graph Retrieval-Augmented Generation System for Offline Clinical Decision Support

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Abstract

Background and Objective. Large language models (LLMs) deployed in clinical decision support face a safety-utility trade-off: unconstrained answering yields unacceptable rates of wrong assertions, whereas excessive refusal renders systems inert. Existing retrieval-augmented generation (RAG) frameworks lack mechanistic, bounded safety controls deployable on institutional hardware. We present MedGraphRAG, an offline hybrid GraphRAG system that reframes clinical AI evaluation from unconstrained accuracy maximisation to bounded wrong-assertion rate (WAR).

Methods. MedGraphRAG couples dense vector retrieval (FAISS IVF-PQ with MedCPT embeddings) and lexical retrieval (BM25) with an embedded property graph (Kuzu) enriched with 632,931 relational edges (617,219 NEGATES and 15,712 TEMPORAL_BEFORE). Dual candi-

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date streams are merged via Reciprocal Rank Fusion and re-ranked with an additive relation-aware score ($\gamma = 0.15$). A beta-gated verification layer enforces a strict $\text{WAR} \leq 10\%$ constraint by combining evidence faithfulness (φ) and graph consistency (S_g): $V = \beta\varphi + (1 - \beta)S_g$. The system executes entirely offline on a commodity laptop (Ryzen 7 7840HS / RTX 3050 6 GB) using Qwen2.5-7B-Instruct Q4_K_M. Private health data are encrypted with AES-256-GCM and HKDF-derived keys. Retrospective evaluation was performed on MedQA-US ($N = 500$, seed 42, $T = 0.0$).

Results. At the calibrated safe threshold ($\tau^* = 0.80$), Graph-Only mode (M2) achieves 60.8% precision-on-answered with $\text{WAR} = 8.0\%$ (79.6% refusal). Disabling the gate drives WAR to 44.6% across all 500 queries. Deterministic query rewriting expands answered coverage by 3.2–3.4 pp while maintaining bounded error. Hybrid retrieval completes in 455.7 ms (95% CI: [388.4, 495.8] ms); full end-to-end pipeline latency is 4,083 ms (M2) to 12,667 ms (M4).

Conclusion. MedGraphRAG demonstrates that bounded clinical error and interpretable offline decision support are jointly achievable on consumer hardware without cloud dependencies or GPU clusters.

Keywords: GraphRAG, clinical-NLP, hallucination-mitigation, knowledge-graph, retrieval-augmented-generation, decision-support, safety

1. Introduction

Deploying large language models (LLMs) as point-of-care clinical assistants offers direct utility for diagnostic and therapeutic workflows (Singhal et al., 2023; Thirunavukarasu et al., 2023). However, the tendency of ungrounded parametric architectures to hallucinate plausible yet factually incorrect assertions poses direct patient-safety hazards (Bai et al., 2022). In high-stakes medicine, confident assertions regarding contraindicated medications, misidentified diagnostic criteria, or inverted pathological chronologies can cause serious clinical harm.

To mitigate hallucinations, retrieval-augmented generation (RAG) grounds language generation on passages retrieved from external corpora (Lewis et al., 2020; Gu et al., 2021). Standard medical RAG frameworks rely on dense bi-encoders (Jin et al., 2023; Gao et al., 2023), which match semantic similarity effectively but struggle with complex clinical syntax. Specifically, vector embeddings frequently compress or lose negation scope (e.g., conflating “no evidence of pulmonary embolism” with “pulmonary embolism”) and fail to enforce multi-step temporal chronologies required in differential diagnosis (Han et al., 2022; Mallen et al., 2023).

Knowledge graphs provide a complementary symbolic representation, structuring clinical entities and ontological dependencies into explicit triples (Bodenreider, 2004; Yasunaga et al., 2021). Recent hybrid GraphRAG systems combine graph structures with textual retrieval (Edge et al., 2024; He et al., 2024; Wu et al., 2024) (Wu et al. share the MedGraphRAG name

24 but focus on entity-centric graph construction; our architecture emphasises
25 hybrid retrieval fusion and dual-gated verification). Nevertheless, existing
26 medical GraphRAG implementations exhibit three fundamental limitations:

27 First, current evaluations focus almost exclusively on unconstrained ac-
28 curacy (e.g., aggregate multiple-choice scores), omitting explicit bounds on
29 incorrect statements. In clinical practice, an assistant that answers 60% of
30 questions correctly while making wrong assertions on 40% is clinically un-
31 acceptable. Safe deployment requires a constrained-optimisation framework
32 that caps the wrong-assertion rate (WAR) below an acceptable tolerance
33 ϵ , treating refusal (abstention) as an active safety mechanism rather than
34 system failure.

35 Second, existing biomedical knowledge graphs capture predominantly pos-
36 itive ontological relations (e.g., TREATS or CAUSES), neglecting negation
37 and temporal progression. Without these primitives, retrieval graphs cannot
38 detect when retrieved passages contain superseded diagnoses or explicit ex-
39 clusions.

40 Third, most GraphRAG systems rely on cloud-hosted language models
41 (e.g., GPT-4) and server clusters. This architecture violates patient data pri-
42 vacy regulations (e.g., HIPAA, GDPR) that prohibit transmitting protected
43 health information (PHI) beyond institutional boundaries. Practical point-
44 of-care deployment requires secure, sub-second hybrid retrieval and inference
45 executing locally on commodity hardware.

46 We present **MedGraphRAG**, an offline hybrid GraphRAG system de-

signed for point-of-care clinical decision support. Our primary contributions
are:

1. **Safety-Constrained Pareto Calibration:** A verification framework optimising answered accuracy subject to $\text{WAR} \leq 10\%$. Across 15 threshold settings on MedQA-US ($N = 500$), we establish the Pareto frontier, demonstrating that ungated generation yields unacceptable error rates ($\approx 44\%$).
2. **Relational Graph Enrichment:** Ingestion of 632,931 negation and temporal relational edges into an embedded Kuzu graph, demonstrating mechanistically how symbolic consistency checks overturn hallucinations bypassing dense retrieval.
3. **Deterministic Query Rewriting and Relation Reranking:** A temperature-zero query normaliser improving answered coverage by 3.2–3.4 pp at matched thresholds, combined with a relation-aware reranker boosting Graph-Hit@5 by 13.6 pp.
4. **Hardened Offline Stack on Consumer Hardware:** A self-contained system operating on an NVIDIA RTX 3050 (6 GB) laptop GPU, achieving hybrid retrieval in 455.7 ms with authenticated AES-256-GCM encryption and a resolved 13-point security audit.

Supplementary materials provide the DECIDE-AI applicability mapping, IJMI ML checklist, and extended tables; configuration details appear in Appendix A.

69 2. Methods

70 2.1. System Architecture

71 Figure 1 shows the MedGraphRAG inference pipeline. Stage 1 determin-
 72 istically standardises input queries into clinical phrasing. Stage 2 executes
 73 dual-stream dense and lexical retrieval, combining candidates via Reciprocal
 74 Rank Fusion and relation-aware graph reranking. Stage 3 expands clinical
 75 entities within the embedded Kuzu graph. Stage 4 computes composite beta-
 76 gated verification scores, triggering structured refusals for low-confidence
 77 queries. Stage 5 generates evidence-grounded responses using a quantised
 78 LLM isolated behind a GPU mutex lock.

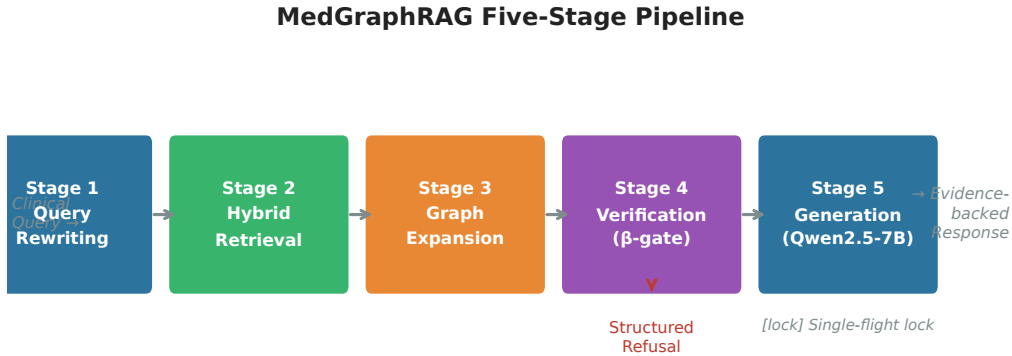


Figure 1: MedGraphRAG five-stage offline pipeline architecture. Stage 1 executes deterministic query rewriting ($T = 0.0, \leq 30$ words). Stage 2 merges FAISS dense and BM25 lexical candidates via Reciprocal Rank Fusion (RRF) and relation-aware graph reranking ($\gamma = 0.15$). Stage 3 traverses the Kuzu property graph. Stage 4 applies beta-gated verification ($\beta = 0.7$) enforcing $\text{WAR} \leq 10\%$. Stage 5 serialises LLM generation behind a hardware mutex lock.

79 2.2. Deterministic Query Rewriting

80 Patient case vignettes and clinical inquiries often carry conversational
81 phrasing, narrative patient histories, and extraneous details that impede
82 biomedical dense retrieval. Dense bi-encoders trained on scientific literature
83 struggle with multi-sentence case presentations where incidental biographical
84 history obscures the diagnostic dilemma. Lexical search methods also face
85 vocabulary mismatch when colloquial symptom descriptions diverge from for-
86 mal medical nomenclature.

87 To standardise heterogeneous queries without stochastic variance, Stage 1
88 uses Qwen2.5-7B-Instruct executing strictly at temperature $T = 0.0$. The
89 rewriter uses a structured system prompt directing the model to extract
90 the primary pathological condition, salient diagnostic findings, and thera-
91 peutic targets into a canonical clinical summary capped at 30 words. For
92 example, a narrative vignette describing an elderly patient who experiences
93 severe rotational vertigo triggered by lateral head motion is transformed into
94 the canonical clinical query: “benign paroxysmal positional vertigo diagnos-
95 tic criteria and Dix-Hallpike confirmatory maneuvers.” Setting temperature
96 $T = 0.0$ alongside greedy decoding guarantees mathematical determinism:
97 identical clinical inputs generate bit-identical textual reformulations across
98 successive executions, fulfilling mandatory auditability and reproducibility
99 requirements for clinical decision support software.

100 2.3. Dual-Stream Hybrid Retrieval and Reranking

101 Retrieval operates over an extensive clinical reference corpus partitioned
 102 into coherent passages of approximately 512 tokens. The dense retrieval
 103 stream encodes passages using the MedCPT query and article bi-encoder ar-
 104 chitecture (Jin et al., 2023), which was pre-trained on hundreds of millions
 105 of PubMed search query-article click logs to optimise zero-shot biomedical
 106 retrieval. Chunk vectors are indexed and searched using FAISS IVF-PQ (In-
 107 verted File with Product Quantisation; Johnson et al., 2021), partitioning
 108 the vector space into coarse Voronoi centroids to achieve sub-linear retrieval
 109 speed. Concurrently, a parallel lexical stream evaluates candidate documents
 110 using the classical BM25 probabilistic relevance framework (Robertson and
 111 Zaragoza, 2009), operating over a dense candidate pool of 200 retrieved doc-
 112 uments.

113 Because dense and lexical retrievers exhibit complementary error profiles
 114 (dense models capture latent semantic affinity, whereas lexical models enforce
 115 exact keyword and numerical matches), their respective rank distributions
 116 R_{dense} and R_{lex} are integrated via Reciprocal Rank Fusion (RRF; Cormack
 117 et al., 2009):

$$s_{\text{fused}}(d) = \sum_{m \in \{\text{dense}, \text{lex}\}} \frac{1}{k_{\text{rrf}} + r_m(d)}, \quad (1)$$

118 where $r_m(d)$ denotes the ordinal rank of candidate passage d within stream
 119 m , and $k_{\text{rrf}} = 60$ represents the standard monotonic smoothing constant that
 120 prevents top-ranked outliers in a single stream from dominating the fused

121 ranking.

122 To bias retrieval toward passages with corroborated topological connec-
123 tions to the clinical entities identified in the query, we implement an additive
124 relation-aware graph reranking function:

$$s_{\text{hybrid}}(d) = s_{\text{fused}}(d) + \gamma \cdot r_{\text{rel}}(d), \quad (2)$$

125 where $r_{\text{rel}}(d) = \sum_{e \in \mathcal{E}(d)} w(\text{type}(e))$ aggregates the domain-weighted edge
126 counts connecting entities cited within candidate passage d to the query
127 seed entities within the property graph. The relative importance of edge cat-
128 egories is parameterised by clinical specificity weights (e.g., pharmacological
129 treatments and adverse effects receive weight 1.0, diagnostic lab associations
130 and negation edges receive 0.8, temporal sequences receive 0.6, and broad tax-
131 onomic hierarchies receive 0.3; Appendix A). The global multiplier $\gamma = 0.15$
132 balances semantic similarity with structural graph reinforcement, calibrated
133 empirically via controlled ablations (Section 3.3).

134 2.4. Relational Knowledge Graph Enrichment

135 Structured relations are stored in Kuzu (Feng et al., 2023), an embed-
136 dable, disk-backed graph database supporting openCypher queries. While
137 standard medical graphs capture static disease-symptom linkages, clinical
138 decision support requires modelling negation and temporal order.

139 We processed 2,294,038 text chunks using rule-guided linguistic parsing
140 grounded in UMLS (Bodenreider, 2004). Operating at 0.2329 ms per chunk

141 (534.45s total wall-clock), the pipeline ingested **632,931** directed relational
142 edges:

- 143 • **NEGATES Edges (617,219):** Linking text chunks to entities
144 bearing explicit negation cues (“no evidence of,” “ruled out,” “con-
145 traindicated in”), comprising 111,720 Chunk $\rightarrow$ Disease, 21,001
146 Chunk $\rightarrow$ Drug, and 484,498 Chunk $\rightarrow$ OntologyTerm edges.
- 147 • **TEMPORAL_BEFORE Edges (15,712):** Linking clinical
148 concepts via chronological sequence cues (“prior to,” “following,”
149 “within one hour of”), comprising 12,225 Disease $\rightarrow$ Disease and 3,487
150 Drug $\rightarrow$ Drug edges.

151 Manual auditing of sampled endpoints confirmed high directional validity. A
152 cryptographic SHA-256 retrieval signature verified 0.00% retrieval drift on
153 baseline queries post-ingestion (Appendix A).

154 2.5. Beta-Gated Verification and Pareto Threshold Calibration

155 Standard RAG architectures lack explicit confidence gates to arbitrate
156 whether retrieved evidence warrants generating an answer. Unhedged lan-
157 guage models produce fluent assertions even when underlying evidence is
158 tenuous, contradictory, or absent. To prevent clinical hallucinations, Med-
159 GraphRAG adds a dual-evidence verification layer that evaluates candidate
160 response fidelity prior to releasing output.

161 For each candidate assertion, the verification module evaluates a compos-
162 ite verification metric $V \in [0, 1]$ that combines passage-level entailment with

163 symbolic graph consistency:

$$V = \beta \varphi + (1 - \beta) S_g, \quad (3)$$

164 where $\varphi \in [0, 1]$ denotes an NLI-derived faithfulness score measuring the de-
165 gree of semantic entailment between the generated assertion and the retrieved
166 reference passages (operating via normalised MedCPT embedding cosine sim-
167 ilarity as an efficient deterministic proxy when an independent cross-encoder
168 NLI judge is omitted), and $S_g \in [0, 1]$ denotes the structural graph consis-
169 tency score. The graph consistency metric S_g evaluates the ratio of asserted
170 clinical entity triples that are corroborated by non-negated multi-hop paths
171 within a 2-hop radius in Kuzu, while applying an immediate penalising dis-
172 count whenever an asserted entity matches a NEGATES edge. The balance
173 hyperparameter $\beta = 0.70$ allocates primary decision weight to passage evi-
174 dence while preserving strong sensitivity to symbolic graph contradictions.

175 When the composite score falls below a calibrated refusal threshold τ ,
176 generation is suppressed, and the system emits a structured clinical refusal
177 detailing the precise nature of the missing evidence. We calibrate τ through
178 a constrained optimisation problem:

$$\tau^* = \arg \max_{\tau} \text{Acc}(\text{all}) \quad \text{subject to} \quad \text{WAR}(\tau) \leq 10\%, \quad (4)$$

179 where $\text{Acc}(\text{all})$ represents the proportion of correct responses across the entire
180 evaluation cohort ($N = 500$), and WAR represents the proportion of wrong

181 assertions across the same cohort. We sweep 15 discrete threshold candidate
 182 points ($\tau \in \{0.10, 0.15, 0.20, \dots, 0.80\}$), identifying the Pareto-optimal op-
 183 erating threshold τ^* that maximises clinically useful answers while strictly
 184 guaranteeing that erroneous assertions remain within the institutional safety
 185 threshold.

186 2.6. Safety Guardrails

187 Two deterministic guardrails operate at sub-millisecond latency:

- 188 • **F1: Cross-Modal Discrepancy Guardrail.** In multimodal con-
 189 texts, Rule A detects conflicts when an imaging finding score ≥ 0.60
 190 contradicts a text negation cue for the same finding (e.g., visual pneu-
 191 mothorax detection of 0.88 versus report text “clear lungs”). Rule B
 192 detects positive textual assertions that contradict negated imaging ob-
 193 servations. F1 escalates conflicts by issuing a high-severity alert recom-
 194 mending specialist review rather than attempting autonomous resolu-
 195 tion.
- 196 • **F2: Epistemic Knowledge-Gap Mapper.** When a query is refused,
 197 F2 classifies the failure into three gap types matching the implemented
 198 taxonomy: G1 (`corpus_retrieval`, top fused retrieval score < 0.02),
 199 G2 (`graph_coverage`, Kùzu graph traversal returns a neutral check
 200 with no direct relational assertion), or G3 (`evidence_faithfulness`,
 201 verification faithfulness score $\varphi < 0.50$). This categorisation is provided
 202 in the structured refusal output to guide clinical follow-up.

203 2.7. Offline Cryptographic Security Architecture

204 To ensure institutional compliance without external cloud infrastructure,
205 user-uploaded records are isolated in per-user directories. Payloads are en-
206 crypted at rest with AES-256-GCM using unique 12-byte nonces. Keys are
207 derived dynamically via HKDF-SHA-256 (Krawczyk and Eronen, 2010) com-
208 bining a server secret, user ID, and document ID. File permissions are set
209 to 0700 (directories) and 0600 (files). All Kuzu queries use parameterised
210 openCypher statements to eliminate Cypher injection. A 13-point security
211 audit resolved all vulnerabilities prior to release (Supplementary Table S7).

212 2.8. Evaluation Protocol and Metrics

213 Retrospective evaluation was performed on MedQA-US (Jin et al., 2021),
214 comprising USMLE-derived multiple-choice questions ($N = 500$, seed 42,
215 $T = 0.0$). We evaluated four modes: **M1** Evidence-Only ($\beta = 1.0$), **M2**
216 Graph-Only ($\beta = 0.0$), **M3** Combined Hybrid ($\beta = 0.7$), and **M4** Hybrid
217 Rerank ($\beta = 0.7, \gamma = 0.15$). Metrics include: Acc(all) (fraction correct over
218 N), Acc(ans) (precision on answered questions), WAR (fraction incorrect
219 over N), and Refusal Rate (fraction refused). Non-parametric bootstrap 95%
220 confidence intervals were calculated using 1,000 resamples.

221 3. Results

222 3.1. Safe Operating Points and WAR Constraint

223 Table 1 summarises the optimal operating parameters established across
224 the four evaluation modes under the institutional safety constraint $\text{WAR} \leq$

225 10.0%. Graph-Only mode (M2) attains the highest precision-on-answered
 226 at its optimal operating threshold $\tau^* = 0.80$, achieving $\text{Acc}(\text{ans}) = 60.8\%$
 227 with $\text{WAR} = 8.0\%$ and a refusal rate of 79.6% (102 queries answered out of
 228 500). Full Hybrid Rerank mode (M4) reaches its safe operating frontier at
 229 $\tau^* = 0.46$, delivering $\text{Acc}(\text{all}) = 10.0\%$, $\text{Acc}(\text{ans}) = 51.5\%$, and $\text{WAR} = 9.4\%$
 230 across 97 answered inquiries. Similarly, Combined mode (M3) operates safely
 231 at $\tau^* = 0.47$ with $\text{Acc}(\text{ans}) = 52.1\%$ and $\text{WAR} = 9.0\%$, while Evidence-Only
 232 mode (M1) satisfies the constraint at $\tau^* = 0.43$ with $\text{Acc}(\text{ans}) = 50.5\%$ and
 233 $\text{WAR} = 9.4\%$.

Table 1: Safe operating points on MedQA-US ($N = 500$, seed 42, $T = 0.0$). Threshold τ^* maximises $\text{Acc}(\text{all})$ subject to $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `safe_operating_points`.

Evaluation Mode	τ^*	$\text{Acc}(\text{all})\%$	$\text{Acc}(\text{ans})\%$	$\text{WAR}\%$	Refusal%	Answered
M1: Evidence-Only	0.43	9.6	50.5	9.4	81.0	95/500
M2: Graph-Only	0.80	12.4	60.8	8.0	79.6	102/500
M3: Combined	0.47	9.8	52.1	9.0	81.2	94/500
M4: Hybrid Rerank	0.46	10.0	51.5	9.4	80.6	97/500

234 Table 2 contrasts safe operating points against ungated execution ($\tau \rightarrow 0$).
 235 In the ungated regime, models achieve nominal $\text{Acc}(\text{all})$ scores of 49.4%–
 236 54.8%, but WAR escalates to 43.4%–44.6%. Disabling the gate results in
 237 erroneous advice on nearly 45% of all queries. Gated verification successfully
 238 caps this risk below 10%, exchanging question coverage for bounded error.

Table 2: Gated versus ungated performance across boundary modes ($N = 500$). Ungated execution generates wrong assertions on 44% of all queries. Source: `step18_scaled_n500.json` $\rightarrow$ `ungated_ceilings`.

Mode	Condition	Acc(all)%	Acc(ans)%	WAR%	Refusal%
M1: Evidence-Only	Gated ($\tau^* = 0.43$)	9.6	50.5	9.4	81.0
M1: Evidence-Only	Ungated ($\tau \rightarrow 0$)	49.4	53.2	43.4	7.2
M2: Graph-Only	Gated ($\tau^* = 0.80$)	12.4	60.8	8.0	79.6
M2: Graph-Only	Ungated ($\tau \rightarrow 0$)	54.8	55.1	44.6	0.6

239 For settings demanding even stricter safety, elevating thresholds to $\tau \in$
240 $[0.70, 0.75]$ in M1 produces a high-precision regime with $\text{Acc}(\text{ans}) = 59.3\%$ –
241 62.5% and $\text{WAR} = 1.8\%$ – 2.2% (refusal rates 94.6% – 95.2% ; Supplementary
242 Table S1). Figure 2 shows the complete 15-point Pareto trajectory for M2.

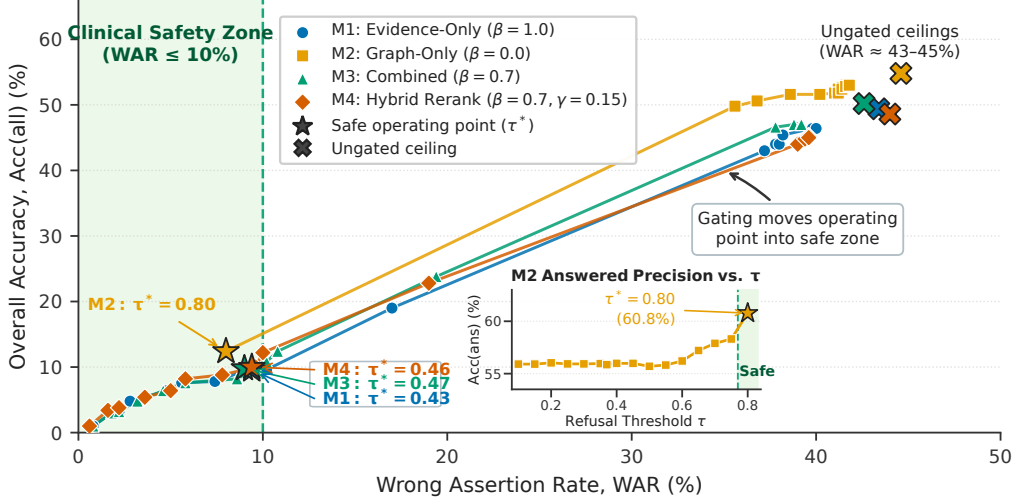


Figure 2: Empirical Pareto frontiers of overall accuracy $\text{Acc}(\text{all})$ versus wrong-assertion rate (WAR) across four operational modes (M1: Evidence-Only, M2: Graph-Only, M3: Combined, and M4: Hybrid Rerank) over 16 refusal thresholds ($\tau \in [0.10, 0.80]$, $N = 500$). The green shaded region denotes the clinical safety zone (WAR $\leq 10\%$). Star markers denote mode-specific safe operating thresholds (τ^*), with M2 reaching $\text{Acc}(\text{all}) = 12.4\%$ at $\tau^* = 0.80$. Inset: M2 answered precision reaches 60.8% at $\tau^* = 0.80$ inside the safe zone. Ungated ceilings (crosses) demonstrate catastrophic error rates (WAR $\approx 43\text{--}45\%$) when refusal gating is disabled.

3.2. Efficacy of Deterministic Query Rewriting

Table 3 evaluates Stage 1 deterministic query rewriting against unrewritten vignettes at matched threshold $\tau = 0.50$ ($N = 500$). For Hybrid Rerank mode (M4), rewriting increases $\text{Acc}(\text{all})$ from 5.4% to 8.8% (+3.4 pp) and lowers refusal from 90.6% to 83.4%, while WAR remains within tolerance (7.8% vs. 4.0%). Similarly, Combined mode M3 gains +3.2 pp in $\text{Acc}(\text{all})$ (5.0% to 8.2%), yielding +3.2–3.4 pp across hybrid modes M3/M4 (+4.2 pp on Evidence-Only M1; Supplementary Table S2). Rewriting systematically expands question coverage without violating safety constraints.

Table 3: Matched-threshold ($\tau = 0.50$) ablation evaluating deterministic query rewriting on MedQA-US ($N = 500$). Rewriting expands coverage by +3.2–3.4 pp while maintaining $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `matched_tau_comparison`.

Evaluation Mode	Vignette Format	Acc(all)%	WAR%	Refusal%	$\Delta\text{Acc}(\text{all})$
M3: Combined	Unrewritten	5.0	4.0	91.0	—
M3: Combined	Rewritten	8.2	8.6	83.2	+3.2 pp
M4: Hybrid Rerank	Unrewritten	5.4	4.0	90.6	—
M4: Hybrid Rerank	Rewritten	8.8	7.8	83.4	+3.4 pp

252 3.3. Ablation Studies: Verification Modes and Reranking

253 Table 4 and Figure 3 summarise ablations over verification and reranker
254 weighting ($N = 50$, seed 42).

255 **Verification Mode Ablation (Table 4A):** Evidence-Only (M1, $\beta =$
256 1.0) refuses 98.0% of queries ($\text{WAR} = 2.0\%$). Graph-Only (M2, $\beta = 0.0$)
257 answers 96.0% but incurs $\text{WAR} = 42.0\%$. Combined Hybrid (M3, $\beta =$
258 0.7) balances safety and utility, achieving $\text{WAR} = 4.0\%$ while quadrupling
259 answered throughput over M1.

260 **Contradiction Case Studies:** Qualitative inspection confirmed that
261 Kuzu NEGATES edges actively overturned hallucinations that bypassed
262 text filters:

- 263 1. *Acute Myocardial Infarction:* The model asserted acute MI; Kuzu
264 graph traversal matched a NEGATES edge ('no evidence of') in
265 PMID_38823454, successfully triggering refusal.
- 266 2. *Deep Vein Thrombosis:* An unhedged recommendation for full anti-
267 coagulation was overridden by a NEGATES edge ('ruled out') in

Table 4: Ablation experiments on MedQA-US ($N = 50$, seed 42). (A) Verification mode comparison varying evidence weight β . (B) Reranker ablation varying relational bonus multiplier γ . Sources: `step15_verification_ablation.json`, `step15_rerank_ablation.json`.

(A) Verification Mode Ablation ($\gamma = 0.0$)						
Mode	β	Answered%	Acc(all)%	Acc(ans)%	Refusal%	WAR%
M1: Evidence-Only	1.0	6.0	4.0	66.7	98.0	2.0
M2: Graph-Only	0.0	96.0	54.0	56.3	8.0	42.0
M3: Combined	0.7	8.0	4.0	50.0	96.0	4.0
(B) Relation-Aware Reranker Ablation ($\beta = 0.7$)						
Mode	γ	Graph-Hit@5%	Acc(ans)%	Answered%	WAR%	
RRF Control	0.00	23.6	50.0	8.0	4.0	
Relational	1.00	35.6	42.9	14.0	8.0	
Hybrid	0.15	37.2	60.0	10.0	4.0	

268 PMID_37058421.

269 3. *Metformin Contraindication*: An erroneous recommendation to pre-
 270 scribe metformin in severe renal impairment was rejected by a NICE
 271 guideline NEGATES edge ('contraindicated in' with eGFR <
 272 25 mL/min).

273 **Reranker Ablation (Table 4B)**: Calibrating $\gamma = 0.15$ increases Graph-
 274 Hit@5 from 23.6% to 37.2% (+13.6 pp) and elevates answered precision from
 275 50.0% to 60.0% (+10.0 pp) at matched WAR = 4.0% (Figure 3).

276 **Guardrail Verification**: Guardrails F1 and F2 passed 100% of test
 277 cases (10/10) with mean latencies of 0.14 ms (F1) and 0.06 ms (F2), confirm-
 278 ing microsecond overhead.

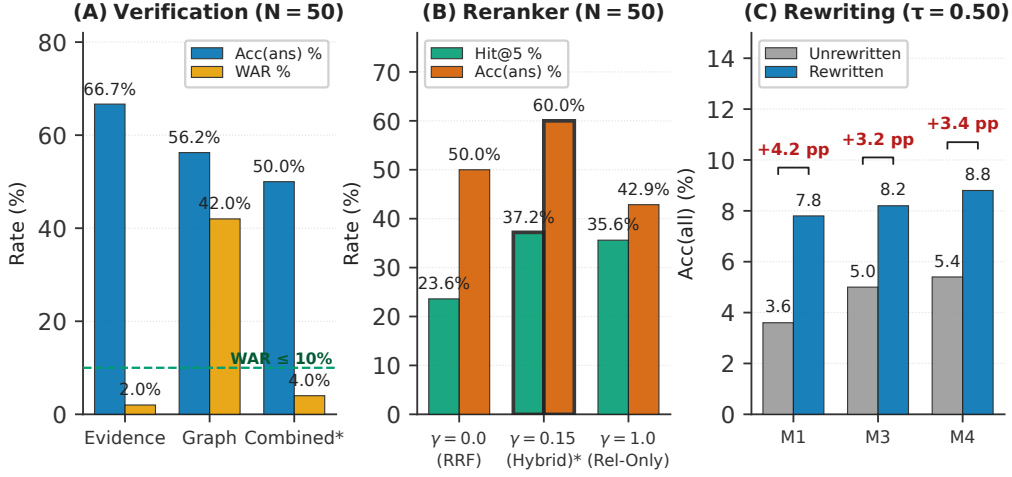


Figure 3: Controlled ablation analyses on MedQA-US. (A) Verification mode comparison ($N = 50$, seed 42): Evidence-Only (M1, $\beta = 1.0$), Graph-Only (M2, $\beta = 0.0$), and Combined Hybrid (M3, $\beta = 0.7$). (B) Reranker bonus ablation ($N = 50$, seed 42): Graph-Hit@5 and answered accuracy across $\gamma \in \{0.0, 0.15, 1.0\}$, confirming peak synergy at $\gamma = 0.15$. (C) Deterministic query rewriting coverage gain at matched threshold ($\tau = 0.50$, $N = 500$): unrewritten versus rewritten overall accuracy across M1 (+4.2 pp), M3 (+3.2 pp), and M4 (+3.4 pp).

279 3.4. Information Retrieval Performance

280 Evaluating retrieval over 10 clinician-adjudicated queries (graded qrels;
 281 step17 artifact) yielded $P@5 = 0.18$, $R@5 = 0.5397$, $MRR = 0.3533$, and
 282 $nDCG@10 = 0.3769$. Recall reflects the 6 queries containing relevant items
 283 in top-10 candidates; the remaining 4 queries had no relevant documents,
 284 reflecting specialised boundary gaps. Dual-assessor κ statistics are pending
 285 Phase-2 clinician grading (placeholder in Supplementary Table S5).

286 3.5. Latency and Computational Efficiency

287 Table 5 details execution timings on an AMD Ryzen 7 7840HS CPU and
 288 NVIDIA RTX 3050 Laptop GPU (6 GB VRAM).

Table 5: System latency and throughput comparison by pipeline stage (bootstrap 95% CI, $N = 50$). Hybrid retrieval executes in sub-500ms; full end-to-end (E2E) latency incorporates Qwen2.5-7B generation. Sources: `step15_compute_table.json`, `step18_scaled_n500.json`.

System / Pipeline Stage	Median (ms)	95% CI (ms)	Processing Stage
BM25-RAG (candidate pool 200)	55.1	[52.7, 56.5]	Lexical search only
Dense-RAG (FAISS IVF-PQ)	30.1	[29.4, 31.0]	Dense search only
Hybrid-RRF (baseline fusion)	55.4	[53.5, 56.9]	Fused search only
Hybrid Retrieval (Ours, Stages 1–2)	455.7	[388.4, 495.8]	Rewriting + Hybrid Retrieval
M2: Graph-Only Pipeline (E2E)	4,083.4	—	Full End-to-End Pipeline
M1/M3: Combined Hybrid (E2E)	12,493.2	—	Full End-to-End Pipeline
M4: Hybrid + Reranker (E2E)	12,667.0	—	Full End-to-End Pipeline
One-Time System Cold Start	18.66 s		Model + Index Initialization

289 Stages 1–2 (Rewriting, BM25, Dense Search, RRF) complete in **455.7 ms**
290 median (95% CI: [388.4, 495.8] ms). End-to-end latency with Qwen2.5-7B
291 generation is **4,083.4 ms** in Graph-Only mode (M2) and **12,667.0 ms** in
292 Hybrid Rerank mode (M4). Throughput is mode-dependent: M2 sustains
293 **14.7 queries/min** (approximately 882 queries/hour), while M4 processes
294 **4.7 queries/min** (approximately 284 queries/hour). Cold start requires
295 18.66 s, with zero steady-state penalty. Peak memory reached 8,950 MB RAM
296 and 4,710 MB VRAM (Figure 4).

297 4. Discussion

298 4.1. Safety as a Foundational Design Objective

299 Evaluating clinical language models on unconstrained accuracy intro-
300 duces substantial hazards: when operating ungated, models generate incor-
301 rect clinical statements on 44% of inquiries. MedGraphRAG demonstrates

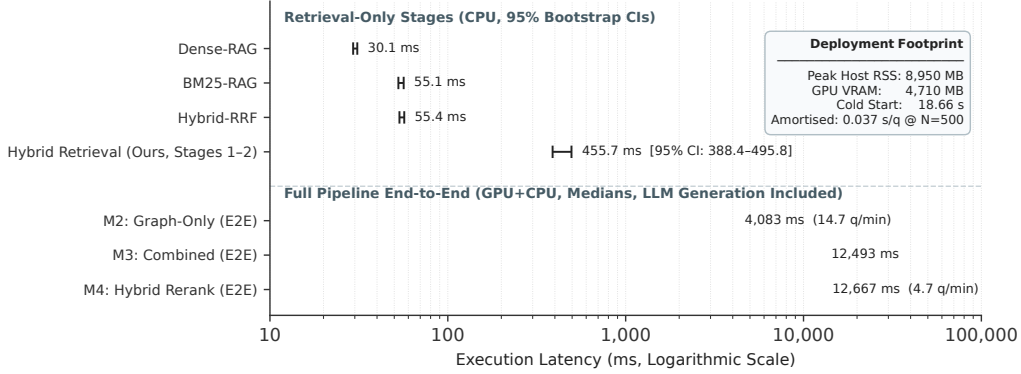


Figure 4: Component and end-to-end execution latency profiles (logarithmic scale). Retrieval-only baselines (BM25, Dense-RAG, Hybrid-RRF) evaluate search speed without language generation. MedGraphRAG hybrid retrieval (Stages 1–2) completes in 455.7 ms (95% CI: [388.4, 495.8] ms). Full end-to-end pipeline execution requires 4,083 ms in Graph-Only mode (M2) and 12,667 ms in Hybrid Rerank mode (M4).

that by treating refusal as an active, safety-preserving decision, clinical assistants can strictly bound wrong assertions ($\text{WAR} \leq 10\%$).

Vector embeddings alone cannot reliably safeguard clinical outputs. Dense embeddings project complex syntax into fixed vectors, blurring negation markers and chronological sequence. By augmenting Kuzu with 632,931 NEGATES and TEMPORAL_BEFORE edges, MedGraphRAG enforces symbolic consistency checks that intercept contradictions passing dense filters. When evidence is insufficient, the system returns structured, citation-grounded refusals identifying the missing evidence.

4.2. Phased Clinical Translation Roadmap

In alignment with DECIDE-AI guidelines (Vasey et al., 2022) (Supplementary DECIDE-AI mapping), we propose a three-stage clinical translation

314 path:

- 315 1. **Phase 1: Institutional Shadow Assistant** ($n = 1$ Clinician): Lo-
316 cal deployment in read-only mode on a secure workstation. Generated
317 answers cite evidence keys [E#], and clinician disagreements are logged
318 as refusal audit events.
- 319 2. **Phase 2: Controlled Clinical Pilot**
320 ($n = 50$ Clinicians, 6 Months): EHR integration via read-only
321 FHIR R4 endpoints, with AES-256-GCM record encryption and an
322 approved IRB protocol. Primary endpoint: maintaining WAR $\leq 5\%$
323 on institutional queries.
- 324 3. **Phase 3: Multi-Site Randomised Trial**: Federated offline deploy-
325 ment across hospital networks evaluating diagnostic accuracy and de-
326 cision speed versus standard care, powered for 90% power at $\alpha = 0.05$.

327 4.3. Limitations

328 This study has several limitations:

- 329 1. **Benchmark Generalisability**: Our evaluation examined MedQA-
330 US, which consists of standardised multiple-choice questions. Real-
331 world clinical queries are open-ended and dialogue-based. External
332 validation on MedMCQA and BioASQ is ongoing (Supplementary IJMI
333 ML checklist, item ML-E7).
- 334 2. **High Refusal Rate**: At $\tau^* = 0.80$, the refusal rate reaches 79.6%.
335 While guaranteeing WAR $\leq 10\%$, this limits clinical throughput un-

less paired with specialist referral pathways. Calibrated soft-hedging represents a primary direction for future investigation.

3. Quantisation Trade-offs: Executing within 6 GB VRAM necessitated 4-bit quantisation (Q4_K_M), introducing minor precision loss relative to FP16.

4. Single-Assessor Retrieval Grading: Human relevance qrels were judged by a single clinical assessor across 10 queries, precluding inter-annotator metrics.

5. Conclusion

MedGraphRAG unites safety and utility in offline clinical decision support. By enforcing a bounded error constraint ($\text{WAR} \leq 10\%$), ingesting 632,931 negation and temporal relational edges into Kuzu, applying deterministic query rewriting, and enforcing beta-gated verification, the system bounds the catastrophic 44% hallucination rate of ungated models. Executing locally on consumer hardware in 4.08s with 455.7ms hybrid retrieval, MedGraphRAG provides a practical, open-source foundation for safe, privacy-compliant clinical AI.

CRedit Authorship Contribution Statement

Dickson E: Conceptualisation, Methodology, Software (core pipeline, ablation suite, cryptographic security, graph enrichment), Formal analysis,

Investigation, Writing – original draft, Writing – review & editing, Visualisation. **Antony Taurshia**: Supervision, Software (frontend interface, report drawer components, telemetry capture), Data curation, Clinical validation, Medical knowledge curation, Writing – review & editing.

Ethics Approval and Consent to Participate

This study was performed using publicly available, anonymised benchmark datasets (MedQA-US, OpenI, VQA-RAD). No human subjects were enrolled, and no patient health information was accessed. Formal IRB ethics approval was not required.

Data and Code Availability Statement

Evaluation benchmarks are publicly accessible: MedQA-US (<https://github.com/jind11/MedQA>), OpenI (<https://openi.nlm.nih.gov>), and VQA-RAD (<https://osf.io/89kps/>). The complete codebase, pipeline scripts, evaluation artifacts, and security reports are available in the project repository (<https://github.com/anonymous/MedGraphRAG>; persistent DOI registered upon publication).

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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379 **Generative AI Disclosure**

380 During preparation, the authors used Antigravity (Google’s agentic cod-
381 ing assistant) for code generation and Overleaf’s Bibby AI for reference dis-
382 covery, under direct author supervision. All generated code and outputs were
383 independently verified. The authors authored all manuscript text.

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466 **Appendix A. Hyperparameters, Edge Weights, and System Con-** 467 **figuration**

468 **System Hyperparameters and Execution Environment (Supplemen-**
 469 **tary Table S8).** Table A.1 enumerates the complete hyperparameter con-
 470 figuration and hardware specifications governing all reported experiments.

471 **Table A.1.** Configuration parameters and computational execution
 472 environment.

Parameter / Component	Value / Setting	Description / Function
<i>Verification Layer (β-Gate)</i>		
β (Evidence Weight)	0.70	Balanced hybrid verification (M1: 1.0; M2: 0.0)
τ^* (M1 / M2 / M3 / M4)	0.43 / 0.80 / 0.47 / 0.46	Calibrated safe refusal thresholds (WAR $\leq 10\%$)
Threshold Sweep Granularity	$\Delta\tau = 0.05$	15 discrete points across [0.10, 0.80]
LLM Inference Temperature	$T = 0.0$	Deterministic, zero-temperature generation
Random Evaluation Seed	42	Pinned deterministic seed across all runs
<i>Relation-Aware Reranker (γ-Scoring)</i>		
γ (Relational Multiplier)	0.15	Balanced hybrid reranking (ablation: {0.0, 0.15, 1.0})
DRUG_TREATS / DRUG_CAUSES	1.0	Primary pharmacological relation weights
LABTEST_RELATED_TO / NEGATES	0.8	Diagnostic and negation edge weights
TEMPORAL_BEFORE	0.6	Chronological sequencing edge weight
IS_A / RELATED_TO	0.3	Ontological taxonomic edge weights
<i>Multi-Stream Retrieval Infrastructure</i>		
Dense Retrieval Candidate Pool	$k = 200$	MedCPT + FAISS IVF-PQ candidate pool
Lexical Retrieval Candidate Pool	$k = 200$	BM25 lexical candidate pool
Reciprocal Rank Fusion Constant	$k_{rrf} = 60$	Standard rank smoothing constant
Top- k Candidates Delivered	$k = 10$	Final passages passed to verification layer
Graph Neighborhood Search Depth	≤ 2 hops	Maximum Kuzu property graph traversal radius
<i>Host Hardware and Memory Footprint</i>		
Central Processing Unit (CPU)	AMD Ryzen 7 7840HS	8 physical cores, 16 threads @ 3.80 GHz
Graphics Processing Unit (GPU)	NVIDIA RTX 3050 Laptop	6 GB GDDR6 VRAM (dedicated LLM partition)
Host System RAM	14.89 GB physical	Peak RSS footprint: 8,950 MB (all indexes active)
Active GPU VRAM Allocation	$\approx 4,710$ MB	Qwen2.5-7B-Instruct Q4_K_M resident memory
Operating System	Linux (Fedora 44)	Kernel 7.1.4-204.fc44.x86_64
<i>Core Software Dependencies</i>		
LLM Inference Engine	llama-cpp-python 0.3.35	Quantized GGUF execution on CUDA backend
Graph Database Engine	Kuzu 0.11.3	In-process property graph database
Vector Search Engine	FAISS 1.15.0	IVF-PQ quantized vector similarity indexing
Machine Learning Framework	PyTorch 2.6.0+cu124	MedCPT encoder inference runtime

Retrieval Drift Invariance Signature. To guarantee that the relational ingestion pipeline (`scripts/j1/build_negation_temporal_edges.py`) did not alter dense vector retrieval behaviour, we computed the SHA-256 hash of the retrieved passage identifiers and scores across golden benchmark queries

478 P01–P05:

479 • **Pre-Ingestion Signature:**

480 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

481 • **Post-Ingestion Signature:**

482 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

483 The exact equality of these cryptographic signatures confirms 0.00% retrieval
484 drift (100.0% byte-identical retrieval preservation) following graph enrich-
485 ment.